Contents

List of Figures

List of Tables

1. Introduction

This report presents the design and simulation of a 42 V to 12 V, 60 W closed-loop synchronous buck converter operating at a switching frequency of 200 kHz. The project focuses on developing a high-efficiency and dynamically stable converter using practical design methodologies and realistic component models.

The converter is designed by considering non-ideal characteristics such as capacitor equivalent series resistance (ESR), inductor DC resistance (DCR), and manufacturer-provided MOSFET models to obtain realistic performance predictions. The design is validated through startup analysis, steady-state operation, output ripple analysis, line regulation, load transient response, efficiency estimation of components and frequency response analysis (FRA) over an input voltage range of 36 V to 48 V.

Initially, a Type-II compensation network was investigated to establish closed-loop operation and evaluate loop dynamics. However, the desired bandwidth and stability margins could not be simultaneously achieved over the entire operating range. Therefore, the compensation network was upgraded to a Type-III compensator, which provides additional phase boost and enables higher crossover frequencies while maintaining adequate phase margin. This resulted in improved transient response, better regulation, and robust closed-loop operation under varying input voltage and load conditions.

The developed simulation model serves as a validated foundation for subsequent hardware implementation using an industrial synchronous buck controller and PCB realization. The project demonstrates practical skills in power-stage design, control-loop compensation, stability analysis, converter modelling, and engineering validation methodologies directly applicable to industrial power electronics product development.

2. Specifications:

2.1 Input voltage (Vin): 42 V
2.2 Output voltage (Vout): 12 V
2.3 Output Current (Iout): 5 A
2.4 Output power (Pout): 60 W
2.5 Switching frequency (Fsw): 200 kHz
2.6 MOSFET: BSC190N12NS3(Infineon)
2.7 Load resistance (R1): 2.4Ω
2.8 Input source resistance: 100mΩ
2.9 Input parasitic inductance: 15nH
2.10 Inductor DCR: 10mΩ
2.11 Capacitor ESR: 30mΩ
2.12 Output capacitor: 22µF
2.13 VCC(V5): 12V
2.14 VREF(V3): PWL (0 0 2m 3.3V)V
2.15 VCC & VREF internal resistance: 10mΩ
2.16 Feedback resistance (Top side): 26.36KΩ

2.17 Feedback resistance (bottom side): 10KΩ
2.18 High side dead time control(B1): V = if((V(PWM_OUT) > 6) & (delay(V(PWM_OUT), 25n) > 6), 9.5, 0)
2.19 Low side dead time control(B2): V = if((V(PWM_OUT) < 6) & (delay(V(PWM_OUT), 25n) < 6), 9.5, 0)
2.20 PWM generation block(B3): V=if(time<0.09ms,0,if(V(VCTRL)>V(VRAMP),9.5,0))
2.21 Error amplifier type: Active Type-II compensation network (R4=10KΩ, R6=20K,C5=1.50nF, C6=62pF).
2.22 Active Type-III Compensation network (R5=1.1KΩ,2.2nF)
2.23 Input decoupling Capacitors (100nf,1µf,10µf)

3. Design Calculations

3.1 Power stage calculation:

- **Duty Cycle(D):** $\frac{Vout}{Vin}$: 12/42 = 0.2857
- **Inductor Selected: 33µH:** $\frac{L(\Delta IL)}{DT}$ = Vin-Vo, L = $\left((42-12)*\frac{0.2857}{1.5*200*1000}\right)$ = 28.57µH. So, nearest standard inductor available is 33µH.)
- **Capacitor selected: 22µF;** ΔVo = $\frac{1}{2}*\frac{T}{2}*\frac{\Delta IL}{2}$: ΔVo = $\frac{\Delta IL}{8fc}$: C = $\frac{\Delta IL}{8f\Delta Vo}$: $\frac{1.5}{8*200*10^{\wedge}3*0.05}$ = 18.75µF. So, nearest standard capacitor is 22 µF. Here 0.05V because we want ripple < 50mv.
- **High side and low side MOSFET selected:** BSC190N12NS3(Infineon): (Imax > 8.6A (5.75 * 1.5 (Safe Margin): 8.6A under typical board-limited thermal condition), (Vds > 42*2(Saf. Margin): 120V), (Rds(on) < 25mΩ to reduce conduction loss: 19mΩ), (Lower figure of merit (Rds(on) * Qg compactible with 200KHz switching frequency).

3.2 Dynamic Duty Cycle and ON-Time Compensation Under Non-Ideal Conditions:

In an ideal buck converter operating in Continuous Conduction Mode (CCM), the output voltage depends strictly on the ideal duty ratio (D = $\frac{Vout}{Vin}$). However, a physical circuit contains non-ideal parasitic elements that introduce internal voltage drops under a full 5A load. These loss channels include:

- MOSFET Static Drain-Source On-Resistance ($R_{DS(on)}$)
- Inductor Direct Current Resistance (DCR = 10mΩ)
- Capacitor Equivalent Series Resistance (ESR = 30mΩ)
- High-side and low-side switching transition and dead-time body diode losses

If this converter were operated open-loop at the calculated ideal high-side ON-time of 1.428µS, these cumulative energy losses would cause the output voltage to drop significantly below the 12V target.

Because this design implements a closed-loop system, the active Type 3 compensation network continuously monitors the output voltage through the feedback divider network.

When the parasitic elements cause the voltage to drop, the error amplifier automatically scales its control voltage output. This corrective action automatically increases the operational pulse-width modulation (PWM) duty cycle from the ideal theoretical value (28.57%) to a practical operating value of approximately 29.57%.

As a result of this automatic loop regulation, the high-side MOSFET ON-time dynamically settles at 1.4788µs during steady-state simulation (verified via cursor measurements at 9.994ms). This automated tracking successfully stabilizes the converter output at the targeted regulation line, completely overcoming the physical losses of the system without manual user adjustment.

3.3 Behavioral Gate Driver Logic: The complementary power MOSFETs are driven using arbitrary behavioral voltage sources, B1 and B2. These sources implement an active break-before-make dead-time generation network using conditional SPICE logic. This logic prevents cross-conduction (shoot-through), a destructive condition where both high-side and low-side MOSFETs turn on simultaneously and short the input power supply directly to ground.

The behavioral expressions govern the gate voltages based on the binary state of the PWM_OUT comparator node:

3.3.1 High-Side Driver Logic (B1 Expression): V = if((V(PWM_OUT) > 6) & (delay(V(PWM_OUT), 25n) > 6), 9.5, 0)

- **Threshold Detection (V(PWM_OUT) > 6)**: Determines if the raw PWM control loop signal has transitioned to a logic-high state (using a 6V threshold out of a 9.5V peak).
- **Logical AND (&)**: Requires both the instantaneous signal and the delayed signal to be true simultaneously before executing a turn-on command.
- **Delay Function (delay(V(PWM_OUT), 25n) > 6)**: Inspects the history of the PWM node. It checks if the signal was *already* high 25 nanoseconds ago.
- **Output Assignment (9.5, 0)**: If both conditions are met, B1 outputs a **9.5V** gate-drive signal to fully saturate the MOSFET channel. If either condition fails, it instantly drops to **0V** to turn the switch off.
- **Physical Effect**: This creates a **25 ns Turn-On Delay**. When PWM_OUT goes high, the driver forces the MOSFET to wait 25 ns before turning on, ensuring the opposite switch has completely turned off first.

3.3.2 Low-Side Driver Logic (B2 Expression): V = if((V(PWM_OUT) < 6) & (delay(V(PWM_OUT), 25n) < 6), 9.5, 0)

- **Threshold Detection (V(PWM_OUT) < 6)**: Determines if the PWM control signal has transitioned to a logic-low state.
- **Delay Function (delay(V(PWM_OUT), 25n) < 6)**: Checks if the control signal was already low 25 nanoseconds ago.
- **Physical Effect**: This creates a complementary **25 ns Turn-On Delay** for the low-side synchronous switch, preventing it from activating prematurely during high-to-low transitions.

3.3.3 Summary of Operation

- **Instant Turn-Off**: When PWM_OUT changes state, the corresponding active driver drops to 0V instantly because the Boolean condition immediately becomes false.
- **Delayed Turn-On**: The incoming driver is forced to wait exactly 25 ns for the delay history window to catch up before it can drive its output to 9.5V.
- **The Result**: A clean 25 ns dead-time window is embedded into every single switching edge, safely protecting the power stage from cross-conduction under all closed-loop operational dynamics.

3.3.4 Gate-Drive Amplitude Selection

The arbitrary behavioral gate drivers are explicitly programmed to deliver a 9.5 V amplitude pulse rather than an idealized 10 V source. This 9.5 V configuration intentionally models the physical voltage drops present in physical power systems.

In a synchronous buck topology, the high-side gate driver relies on a bootstrap capacitor charged from an internal 10 V supply rail. The current must traverse a physical bootstrap diode, which introduces a standard 0.5 V forward voltage drop, leaving exactly 9.5 V available at the gate. Furthermore, inspecting the BSC190N12NS3 datasheet confirms that the transistor channel achieves full saturation past 8 V, meaning a 9.5 V drive pulse completely minimizes Rds(on) conduction resistance while successfully mitigating unnecessary gate charge losses (Qg) and safeguarding the gate-oxide insulation layer from voltage overstress.

3.4 **PWM Generation and Closed-Loop Control Architecture:** To regulate the converter output precisely at 12V under dynamic line and load conditions, a closed-loop negative feedback structure is modelled. This section details the mathematical generation of the central pulse-width modulation signal (V_{PWM_OUT}) that acts as the timing trigger for the behavioural gate drivers.

3.4.1 Voltage sensing, Bias power and error generation: The controller logic requires a stable auxiliary rail and an absolute reference voltage to process feedback signals accurately:

- **Bias supply voltage (Vcc):** Set to 12V, acting as the power supply for the internal error amplifier and providing the 9.5V amplitude output voltage limits for the behavioral gate signals.

- **Reference voltage (Vref):** Set to a piecewise linear function PWL (0s 0v 2.2ms 3.3v) This parameter maps a highly linear controlled voltage ramp starting from an initial unenergized state (0V at t=0) up to a stable logic threshold of 3.3V at t=2ms, corresponding to a continuous voltage slope(dv/dt) of 1650 V/s. This specific ramp rate is designated to prevent excessive inrush current while establishing a clean startup trajectory for the error amplifier.

The actual output voltage (Vout) is scaled down via a behavioral resistor divider network ration (H = 10K / (10K + 26.36K) = 0.275) to match the reference target (12 * 0.275 = 3.3V). The error generation block continuously calculates the instantaneous tracking deviation:

$$V_{feedback}(t) = 0.275 * V_{out}(t)$$
$$V_{error}(t) = 3.3V - V_{feedback}(t)$$

3.4.2 Feedback voltage divider parameter: To scale the nominal 12V output down to match the internal 3.3V Control loop reference (Vref), an analytical voltage divider derivation is executed. Setting the low-side grounding resistor (R_{low}) to a standard engineering baseline of 10KΩ to balance noise immunity against thermal leakage current, the high-side feedback resistor (R_{high}) is calculated using the standard non-inverting scaling:

$$V_{ref} = V_{out} * \left(\frac{\mathbf{R}low}{Rhigh+Rlow}\right)$$

Rearranging the formula to solve target high-side resistor (R_{high}):

$$R_{high} = R_{low} * \left(\frac{Vout}{Vref} - 1\right)$$

$$R_{high} = 10K * \left(\frac{12}{3.3} - 1\right) = 26.363K\Omega$$

This precise mathematical derivation yields a feedback scaling ratio of H = 0.275, ensuring the error generation block continuously calculates tracking derivations with zero steady-state offset error:

3.4.3 The Behavioral PWM Comparator Logic: The central switching command (V_{PWM_OUT}), is generated by comparing a dynamic control error voltage (V_{CTRL}) against an independent, high-frequency sawtooth ramp waveform (V_{RAMP}) oscillating at the converter's switching frequency (200KHz).

The mathematical condition modelled inside the behavioral PWM block is defined as follows:

$$\text{If } V_{CTRL}(t) > V_{RAMP}(t) = V_{PWM\_OUT} = 9.5V$$

$$\text{If } V_{CTRL}(t) \leq V_{CTRL}(t) = V_{PWM\_OUT} = 0V$$

This translates into a high-frequency square wave the PWM_OUT node, whose pulse width shifts proportionally with the magnitude of V_{CTRL}.

3.4.4 Closed-loop feedback compensator structure: The control voltage(V_{CTRL}) is continuously updated by a proportional-integral(PI) error amplifier network modelled via behavioural voltage expressions. The sensing loop functions through three distinct stages.

- **Voltage sensing:** The actual output voltage (Vout) is scaled via a behavioral feedback ratio(H) to yield Vfeedback(t) = H * Vout(t).

- **Error amplification:** The scaled feedback signal compared against a stable internal DC reference voltage(Vref) to compute the transient error voltage:

$$Verror(t) = Vref - Vfeedback(t)$$

- **PI control action:** The mathematical engine applies proportional and integral gains to generate the instantaneous control signal feeding the PWM stage:

$$VCTRL(t) = Kp * V_{error}(t) + Ki\int_0^t Verror(t)dt$$

3.4.5 Control Loop Compensator Design and Evolution

Step 1: Initial Type-II Analytical Baseline

To begin the control loop design, a standard Type-II compensator structure was analytically calculated as a baseline using an initial power stage target filter of L = 33 uH and C = 22 uF.

The LC resonant frequency is:

$$F_{LC} = \frac{1}{2\pi\sqrt{LC}}$$

Substituting the values: $F_{LC} = \frac{1}{2\pi\sqrt{33*10^{-6}*22*10^{-6}}} = 5.9KHz$

Step 2: Analytical Selection of Compensator Zero

The compensator target zero frequency (Fz1) is selected to boost phase before the crossover region, targeted at roughly 20% of the expected crossover bandwidth (Fc = Fsw / 10 = 20 kHz):

Fz1 = Fc / 5 = 4 kHz

Using the error amplifier feedback path parameters where R6 = 20 kΩ:

Fz1 = 1 / (2 * pi * R6 * Czero)

Rearranging to solve for the zero capacitor:

Czero = 1 / (2 * pi * 20 kΩ * 4 kHz) = 1.989 nF
An initial market-available value of 1.5 nF was selected for the zero-charging path.

Step 3: Analytical Selection of High-Frequency Pole

The high-frequency pole (Fp1) is positioned at half the switching frequency to roll off high-frequency PWM switching noise:
Fp1 = Fsw / 2 = 200 kHz / 2 = 100 kHz
Rearranging the pole equation to solve for the high-frequency filter capacitor:
Fp1 = 1 / (2 * pi * R6 * Cpole)
Cpole = 1 / (2 * pi * 20 kΩ * 100 kHz) = 79.57 pF
A standard market-available value of 62 pF was selected.

Step 4: Simulation Optimization and Transition to Type-III

The analytical Type-II values calculated above (R6 = 20 kΩ, 1.5 nF, and 62 pF) were entered into LTspice as an initial design checkpoint. During dynamic transient and frequency response testing, two critical plant behaviors required a design adjustment:

1. **Filter Scaling**: The bulk output capacitance was increased from 22 uF to 60 uF to reduce steady-state output voltage ripple and improve load transient performance, shifting the physical LC double pole down to 3.58 kHz.

2. **Loop Instability**: Because this is a Voltage-Mode Control converter in Continuous Conduction Mode, the sharp 180-degree phase drop of the double pole caused the Type-II network (which is limited to a 90-degree boost) to exhibit inadequate phase margin. To achieve loop stability, the network was optimized into a Type-3 configuration by adding a secondary pole-zero network (R1 = 1.1 kΩ, C3 = 2.2 nF) across the upper feedback resistor (Rfb_top = 26.36 kΩ). This structural upgrade provides the necessary 180-degree phase boost to safely achieve the final optimized 24.02 kHz crossover frequency and 55.54-degree phase margin.

The final optimized schematic component are:

- **Error Amp Loop**: R6 = 20 kΩ C6 = 1.5 nF, C5 = 62 pF
- **Feedback Passive Divider**: Rfb_top = 26.36 kΩ, Rfb_bot = 10 kΩ
- **Type-III Phase-Boost Branch**: R1 = 1.1 kΩ, C3 = 2.2 nF

3.4.6 Decoupling capacitor selection

A. Total Input Capacitance Calculation

The input stage uses three capacitors placed in parallel across the power rail. The total effective input decoupling capacitance (Cin) is the direct sum of these individual values:

- Cin = C4 + C7 + C8
- Cin = 100 nF + 1 uF + 10 uF
- Cin = 0.1 uF + 1.0 uF + 10.0 uF
- Cin = 11.1 uF (or 11.1 * 10^-6 Farads)

B. Converter Operating Duty Cycle Calculation

The nominal duty cycle (D) is required to calculate input current switching behavior. It is defined by the ratio of the output voltage to the input voltage:

- D = Vout / Vin
- D = 3.3 V / 42 V
- D = 0.07857 (or 7.86%)

C. Peak-to-Peak Input Voltage Ripple Calculation

The high-side MOSFET draws sharp pulses of current from the input rail during turn-on. The peak-to-peak input voltage ripple (Delta Vin) is calculated using the standard buck converter input equation:

- Formula: Delta Vin = (Iout * D * (1 - D)) / (fsw * Cin)

Substituting the circuit parameters (Iout = 1.375 A, D = 0.07857, fsw = 200,000 Hz, Cin = 11.1 * 10^-6 F):

- Step 1 (Numerator): 1.375 * 0.07857 * (1 - 0.07857) = 1.375 * 0.07857 * 0.92143 = 0.09955
- Step 2 (Denominator): 200,000 * (11.1 * 10^-6) = 2.22
- Step 3 (Division): Delta Vin = 0.09955 / 2.22
- Delta Vin = 0.04484 V (or 44.84 mV)

D. Input LC Network Resonant Frequency Calculation

The 15 nH parasitic trace inductance (L2) interacts with the 11.1 uF total input decoupling capacitance, creating a natural low-pass LC filter. Its resonant frequency (Fres) is calculated as follows:

- Formula: Fres = 1 / (2 * pi * sqrt (L2 * Cin))
  Substituting the parasitic components (L2 = 15 * 10^-9 H, Cin = 11.1 * 10^-6 F):
- Step 1(Multiplication): L2 * Cin = (15 * 10^-9) * (11.1 * 10^-6) = 1.665 * 10^-13
- Step 2 (Square Root): sqrt(1.665 * 10^-13) = 4.0804 * 10^-7
- Step 3 (Multiplier): 2 * 3.14159 * (4.0804 * 10^-7) = 2.5638 * 10^-6
- Step 4 (Inversion): Fres = 1 / (2.5638 * 10^-6) = 390,043 Hz
- Fres = 390.04 kHz

E. Summary of Input Network Performance

The calculated mathematical metrics validate the electrical viability of the input power stage network configuration:

- **Ripple Suppression**: The 11.1 uF capacitor array successfully bounds the input voltage ripple to 44.84 mV. This corresponds to an exceptionally low ripple profile of just 0.1% relative to the 42V DC supply bus.
- **Frequency Isolation**: The localized resonant frequency of 390.04 kHz sits safely outside the core 200 kHz switching frequency of the plant. This 190 kHz frequency separation prevents the fundamental converter switching action from exciting the input filter into unstable resonance.
- **Harmonic Q-Factor Damping**: Because the 390.04 kHz resonant point sits near the 400 kHz second switching harmonic, the circuit is vulnerable to parasitic ringing. This hazard is thoroughly mitigated by the 0.1-Ohm internal series resistance (Rser) of the source, which dampens the quality factor (Q) of the LC loop and stabilizes the voltage boundary directly at the high-side MOSFET drain.

4 Circuit overview and diagram:

4.1 Circuit diagram

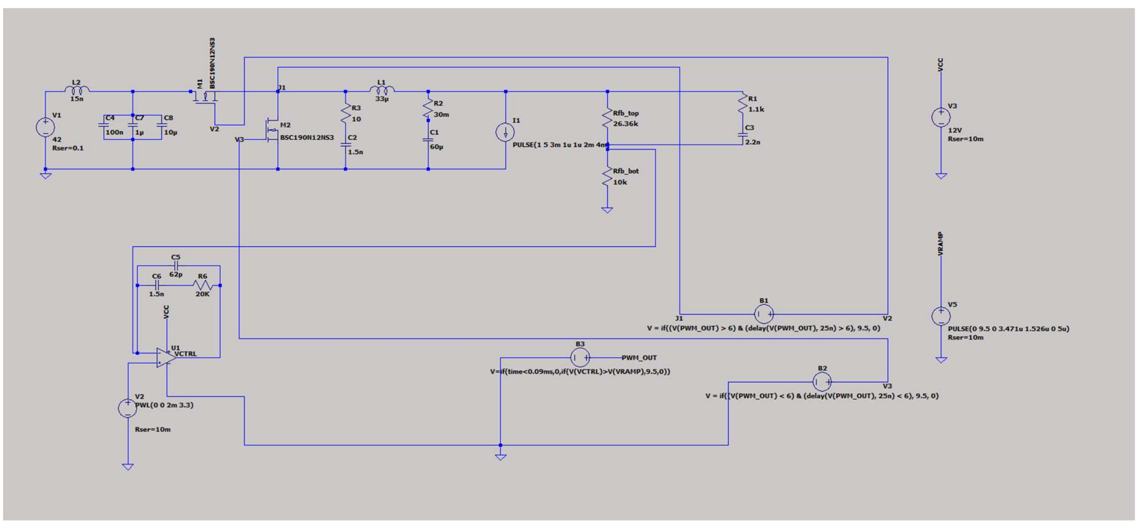


Figure-1: Circuit diagram

4.2 Circuit overview: This circuit is a closed-loop synchronous buck converter simulation schematic utilizing an analog Type-III compensation network. It regulates a high DC input voltage down to a stable, lower DC output voltage through a pulse-width modulation (PWM) feedback loop.

- **Core Stages:**

- **Power Stage**: A DC voltage source (V_1, 42V) supplies power through an LC input filter (L_2, C_4, C_7, C_8).
- **Switching Network**: Two N-channel MOSFETs (M_1 as the high-side switch, M_2 as the low-side synchronous rectifier) drive the main LC output filter (L_1, C_1) and the load resistor (R_4).
- **Feedback & Sensing**: A resistive voltage divider (R_{fb_top}, R_{fb_bot}), samples the output voltage Vout and feeds it to the error amplifier.
- **Type-III Compensator**: Built around operational amplifier U1 combined with resistors (R_6) and capacitors (C_5, C_6), this block adds two poles and two zeros to stable control loop dynamics and ensure rapid transient response.
- **PWM Generation & Control**: Behavioral voltage sources (B_1, B_2, B_3) act as logic gates and drivers. They compare the compensator output (VCTRL) against a ramp signal (VRAMP via source V_5) to dynamically generate complementary PWM gating signals with fixed dead-times for the MOSFETs.

4.3 Circuit Working and Operational Theory

The synchronous buck converter operates by converting a high DC input voltage V1 = 42V to a lower, regulated DC output voltage via controlled high-frequency switching. The operational theory spans three main phases: energy storage, synchronous rectification, and closed-loop regulation.

4.3.1 Energy Storage Phase (M_1 ON, M_2 OFF)

- When the pulse-width modulation (PWM) control signal goes high, the high-side MOSFET M1 turns on while the low-side synchronous MOSFET M2remains off.
- Current flows from the input filter (L2, C4, C7, C8) through (M1) and into the output inductor (L1).
- During this interval, energy is stored in the magnetic field of (L1), and the inductor current increases linearly.
- Simultaneously, current is delivered to the load (R4) and charges the output smoothing capacitor (C1).

4.3.2 Synchronous Rectification Phase (M_1 OFF, M_2 ON)

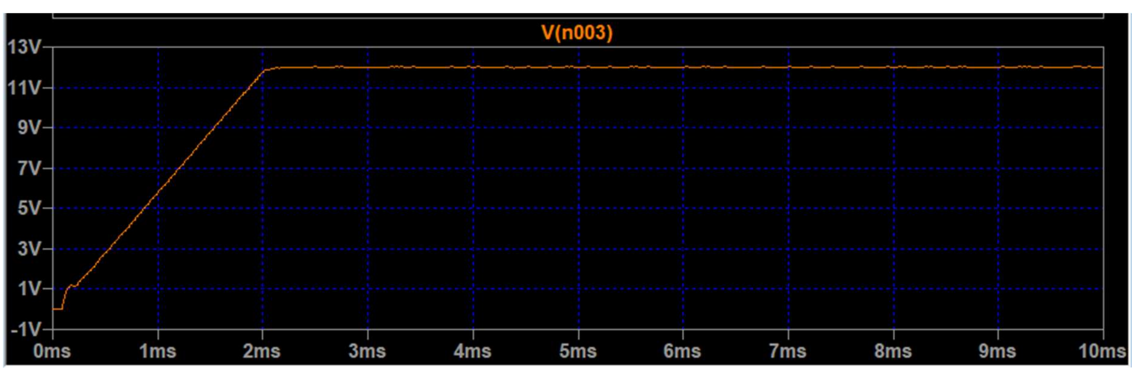
- When the PWM signal transitions low, M1 turns off. To prevent simultaneous conduction (shoot-through), a dead-time delay is introduced before M2 turns on.
- Once M2 conducts, the inductor L1 reverses its voltage polarity to maintain current continuity.
- The stored energy in L1 discharges, and the inductor current decreases linearly, freewheeling through M2.
- Using M2 as a synchronous rectifier instead of a conventional diode significantly reduces conduction losses (I^2 $R_{DS(on)}$), optimizing converter efficiency.

4.3.3 Closed-Loop Regulation and Type-III Compensation

- To maintain a rigid output voltage under varying load conditions, the circuit utilizes a continuous feedback loop:
- **Voltage Sensing:** The output voltage is continuously sampled by the resistive divider (R_{fb_top}, R_{fb_bot}) and fed into the inverting input of the operational amplifier (U1).
- **Error Amplification & Compensation:** The error amplifier compares the sensed feedback voltage against a reference voltage (V2). The Type-III compensation network (C5, C6, R6) along with input components introduces two poles and two zeros. This configuration provides the necessary phase boost to counteract the double-pole resonance caused by the (L1 - C1) output filter, ensuring excellent system stability and rapid transient response.
- **PWM Modulation:** The resulting control voltage (VCTRL) from the compensator is fed to a behavioral voltage source (B3), where it is compared against an independent fixed-frequency sawtooth ramp signal (VRAMP) generated by (V5).
- **Duty Cycle Adjustment:** The intersection of (VCTRL) and (VRAMP) determines the duty cycle (D) of the PWM output signal (VPWM_OUT). If the output voltage drops, (VCTRL) increases, which widens the PWM pulse width (D), forcing more energy into the power stage to restore the target output voltage.

5 Simulation result and analysis:

5.1 Output voltage analysis V(n003): The simulated output voltage varies from peak 12.02V to 11.96V with an average output of 11.99V. This result in a total peak to peak ripple (ΔVout) of ~60mv.



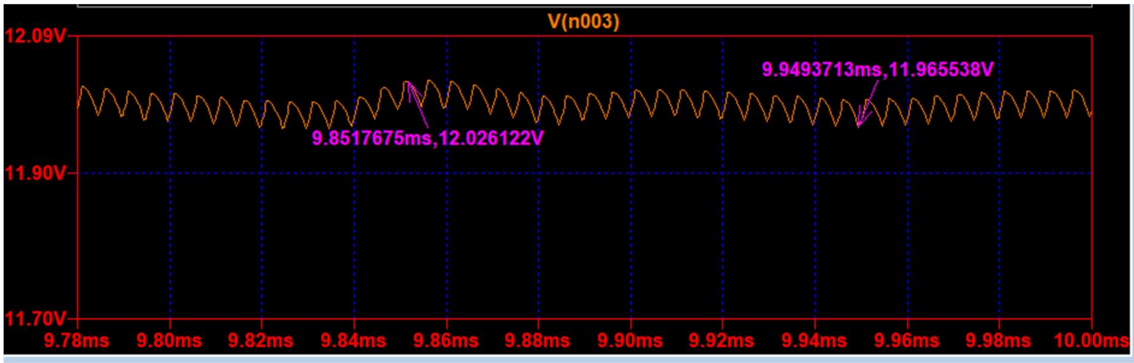


Figure 2: Output voltage

- **Waveform behaviour:** The output voltage is regulated around 12.005. A small high-frequency switching ripple is observed at the switching frequency of 200KHz. In addition, a low-frequency oscillatory component is present due to closed-loop control dynamics and compensator behaviour. This converter remains regulated and stable with an average output voltage close to the target value.

5.2 Inductor current analysis: The Converter operates in continuous conduction mode (CCM). The inductor current varies between approximately 4.316A to 5.732A with an average current of about 4.99A corresponding closely to the rated 5A load condition.

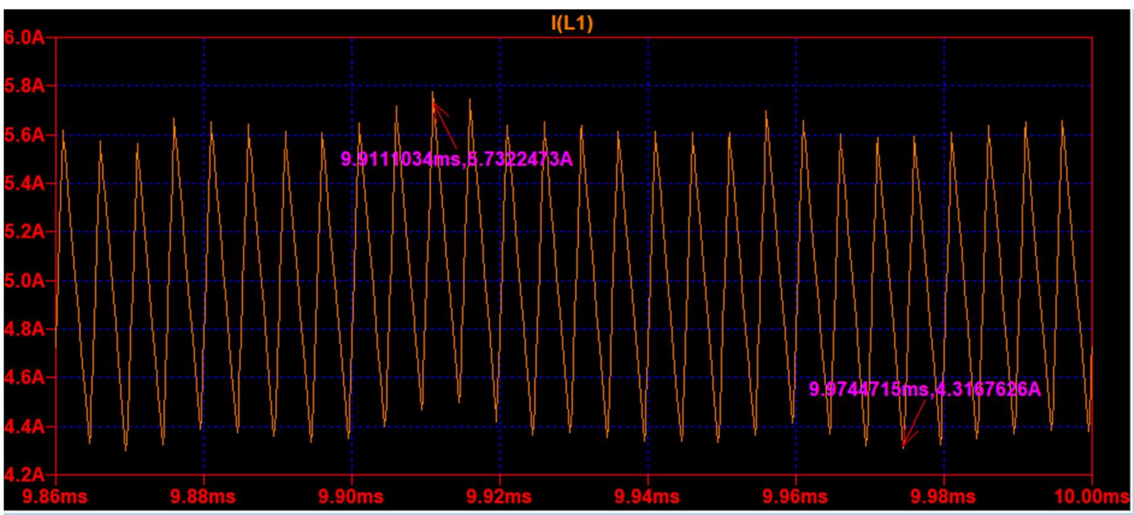


Figure 3: Inductor current

5.3 Gate drive timing verification: The gate-drive waveforms of the high-side V(V2,J1) and low-side MOSFETs V(v3) were analysed to verify proper synchronous switching operation of the converter. Complementary PWM signals were applied to both MOSFETs with a controlled dead-time to prevent simultaneous conduction and shoot-through current. The behavior was observed based on the structural logic of the circuit.

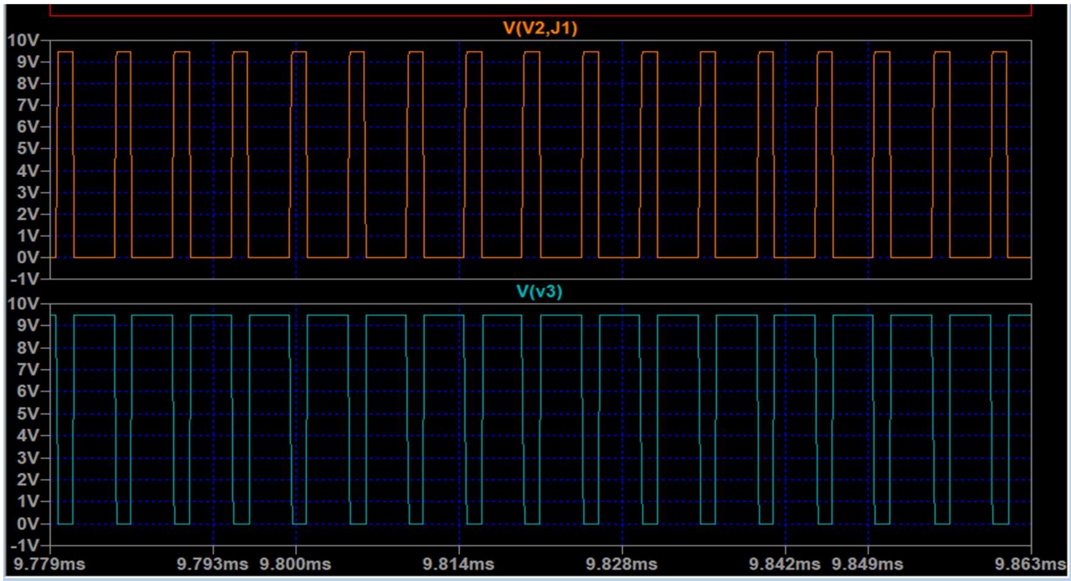


Figure 4: Gate drive timing

5.3.1 Command to Gate propagation delay: When tracking the low side V(V3) against the high-side gate execution V(V2,J1), a discrete transport delay of exactly ~24.258ns was captured via cursor analysis. This perfectly matches the ns delay parameter programmed into arbitrary behavioral source expressions to establish a safe propagation buffer.

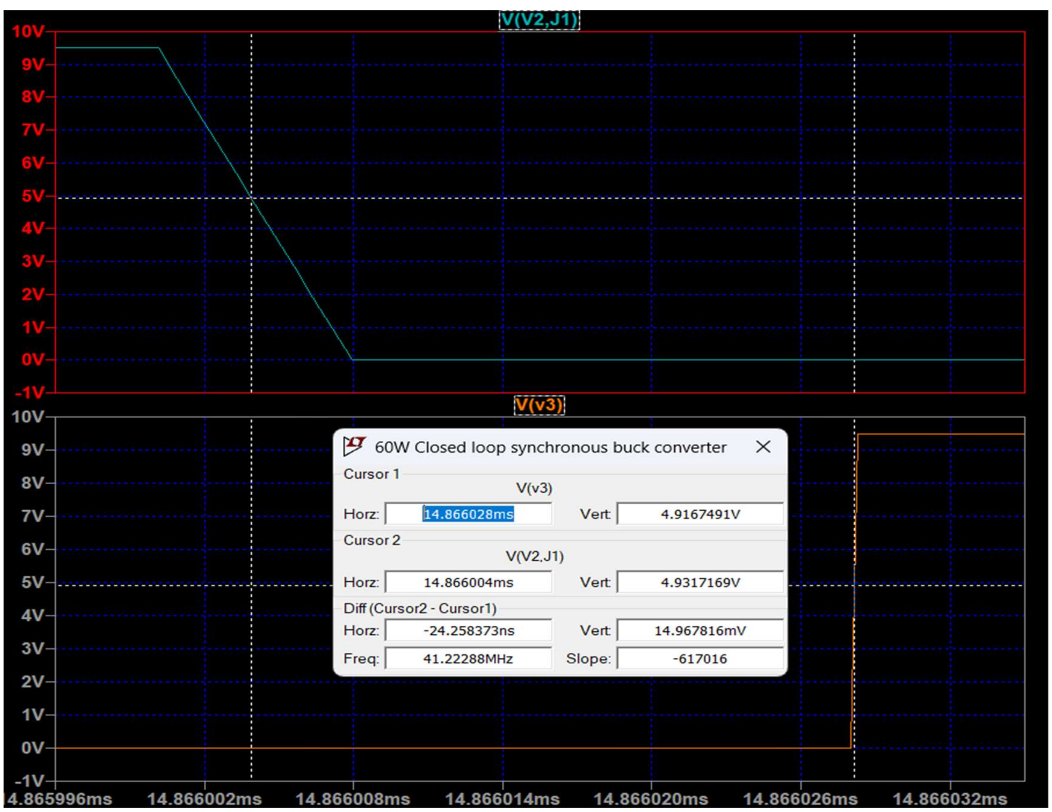


Figure 5: Discrete transport delay

5.4 Switch Node Voltage Analysis

5.4.1 High side gate driver node V(v2): The waveform shows the high-side gate driver voltage V(v2) pulsing rapidly between approximately 0V and 51.31V over a 15ms transient simulation window. The high-density orange blocks indicate continuous, high-frequency Pulse-Width Modulation (PWM) switching, while the thin vertical lines reveal that the closed-loop controller is dynamically adapting the pulse widths to stabilize the converter.

- **Identification of Voltage Levels**
  - **High State (≈ 51.31V):** This represents the active bootstrap or floating gate-drive rail voltage, which supplies enough positive gate-to-source voltage (VGS) to turn the high-side N-channel MOSFET fully ON.
  - **Low State (0V or slightly negative):** This is the deactivated ground/source reference level that completely discharges the gate charge, holding the high-side MOSFET firmly OFF.

- **Closed-Loop Regulation Dynamics**
  - **Transient Duty Cycle Adjustments:** The visible gaps and irregular vertical lines (especially around 1ms to 4ms) show the closed-loop feedback network responding to a transient event (such as a sudden load step or input line change).
  - **Pulse-Width Modulation (PWM):** The loop is modulating the duty cycle D in real time to correct the output error and drive the system back toward a stable steady state.

- **High-Frequency Switching Density**
  - **Continuous Conduction:** The solid orange fill proves that the converter is operating at a high switching frequency (likely hundreds of kHz) with an ultra-dense pulse train.
  - **Mode Verification:** This profile is highly consistent with Continuous Conduction Mode (CCM), maintaining steady current flow throughout the inductor filter network

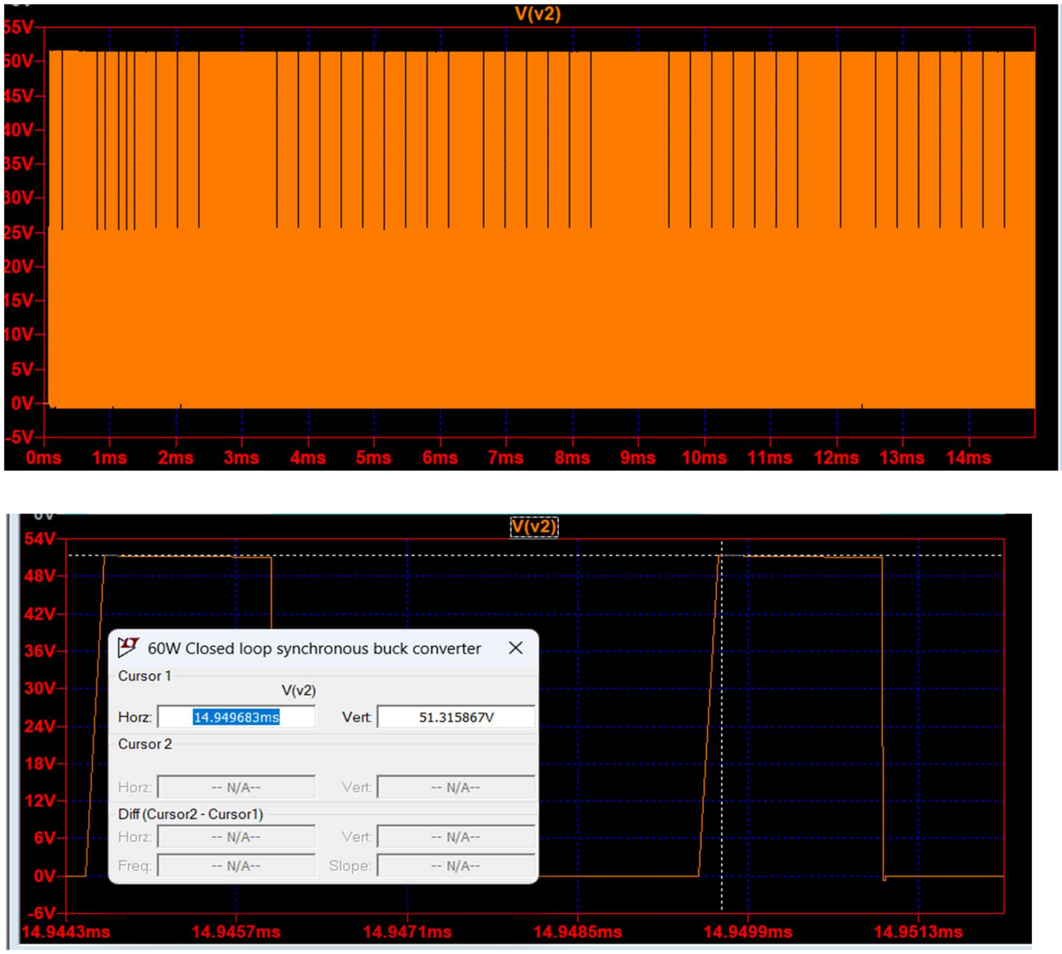


Figure 6: Switch node voltage

5.5 Physical power switch node V(J1): The waveform shows the actual physical power switch node voltage **V(j1)** swinging rapidly between approximately 0V and 42V over a 15ms transient window. Because this is the physical switch node (SW) rather than a isolated gate voltage, the peak value explicitly identifies the converter's DC input rail (Vin ~ 42V). The dense magenta blocks confirm high-frequency switching, while the visible vertical gaps trace closed-loop duty cycle adjustments balancing out transient load or line fluctuations.

➢ **Input Voltage Identification**

- **DC Input Voltage (Vin = 41.83):** The flat upper ceiling of the square wave sits at approximately 41.83V nearly 42V.

- **Physical Context:** This confirms that when the high-side N-channel MOSFET turns on, it directly couples the 42V power input rail to the switch node network.

➢ **Closed-Loop Regulation Footprint**

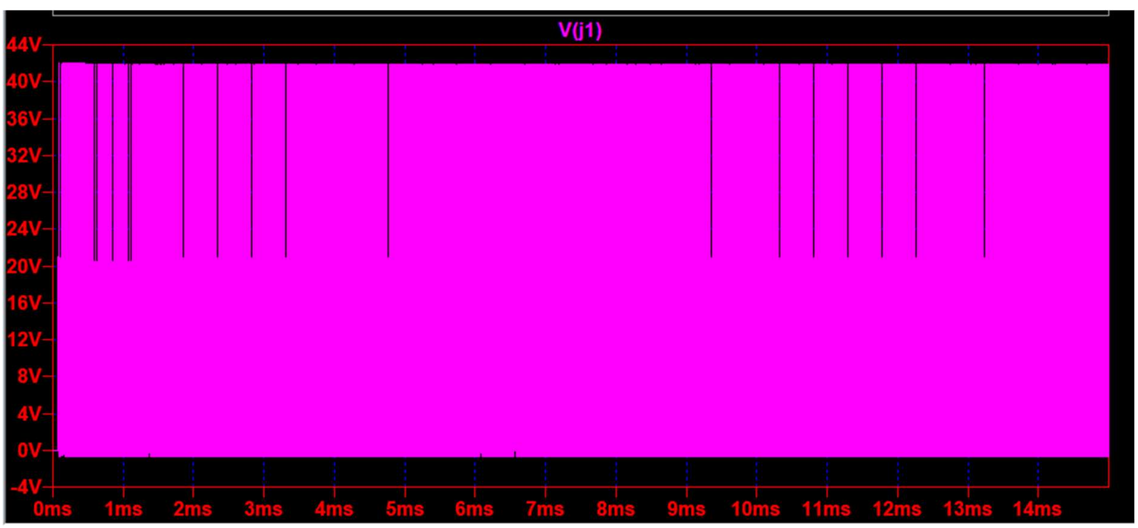
- **Transient Dynamic Adjustments:** Noticeable spacing variations and uneven vertical slots appear distinctly between 0.5ms–2ms, 5ms–6.5ms, and 11ms–14ms.
- **Control Loop Action:** These visual breaks prove that the closed-loop error amplifier is modifying the duty cycle pulse widths in real-time to regulate the system's output voltage through transient disturbances.

➢ **Continuous Conduction Mode (CCM)**

- **Solid Zero-Volt Floor:** The lower envelope rests strictly at or slightly below 0V throughout the entire 15ms timeline without floating up.
- **Filter Context:** This confirms the converter stays solidly within Continuous Conduction Mode (CCM). It proves that inductor current never drops to zero, which would otherwise trigger an intermediate resonant idle period resting at Vout.

➢ **Switching Stress**

- **Power Stress:** The continuous, high-density switching profile confirms steady-state power transfer.
- **Conduction Losses:** The tiny voltage drops visible below 0V on the bottom rail represent current flowing through either the low-side synchronous MOSFET's channel resistance RDS(on) or its parallel body diode during dead-time windows.



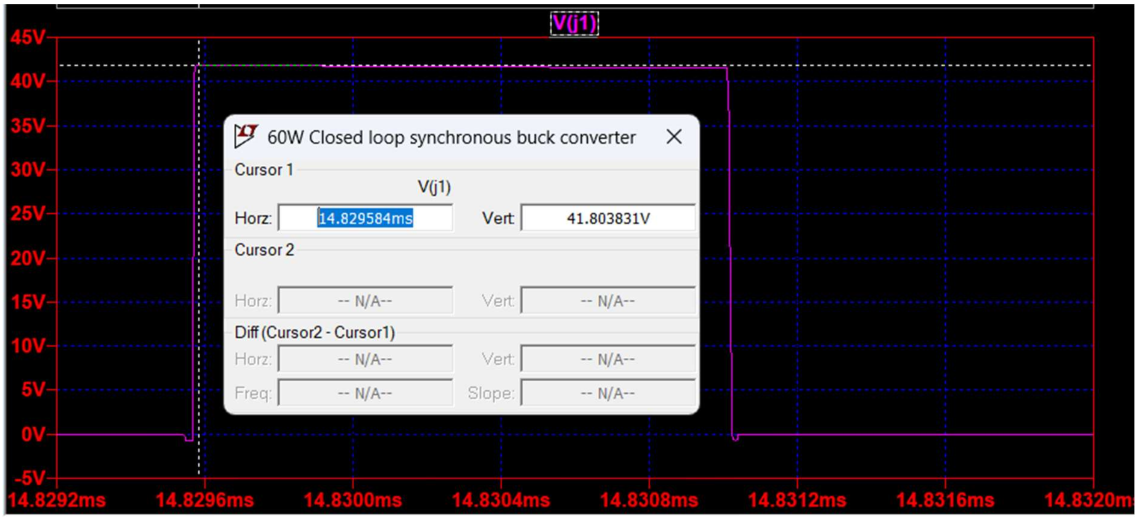


Figure 7: Physical power switch node V(J1)

5.6 Control loop command V(VCTRL): The waveform shows the control voltage V(VCTRL), which is the output of the closed-loop error amplifier (or compensator) fed into the PWM modulator. The waveform exhibits a clear two-phase response: a highly linear startup ramp from 0V to approximately 2.65V during the first 2.2ms, followed by a well-stabilized steady-state operating band (centred around 2.65V) featuring high-frequency switching noise or ripple.

➢ **Startup Transient Phase (0ms to 2.2ms)**

- **Soft-Start Tracking:** The control voltage rises linearly at startup. This indicates the closed-loop controller's soft-start mechanism is active, slowly increasing the duty cycle to prevent massive inrush currents and output voltage overshoot.
- **Slew-Rate Limiting:** The linear slope shows that the error amplifier output is monotonically tracking the internal reference ramp to safely charge the output filter capacitors.

➢ **Closed-Loop Regulation & Steady-State (2.2ms onwards)**

- **Loop Settling Point:** At approximately 2.2ms, the control loop successfully locks onto the target regulation point. The average control voltage settles firmly at ~ 2.65V.
- **Dynamic Equilibrium:** This steady-state value represents the precise control effort required by the error amplifier to maintain the targeted stable output voltage from your 42V input rail.

➢ **High-Frequency Noise & Modulation Band**

- **Control Ripple:** The thick, fuzzy band spanning between 2.45V and 2.90V is caused by high-frequency switching noise, inductor current ripple feedback, or a deliberate slope compensation ramp mixed into the modulator.
- **Stability Verification:** There are no low-frequency oscillations, subharmonic tracking issues, or chaotic hunting behaviours in the steady-state band. This proves that loop compensator (Type III network) has an adequate phase margin and is stable.

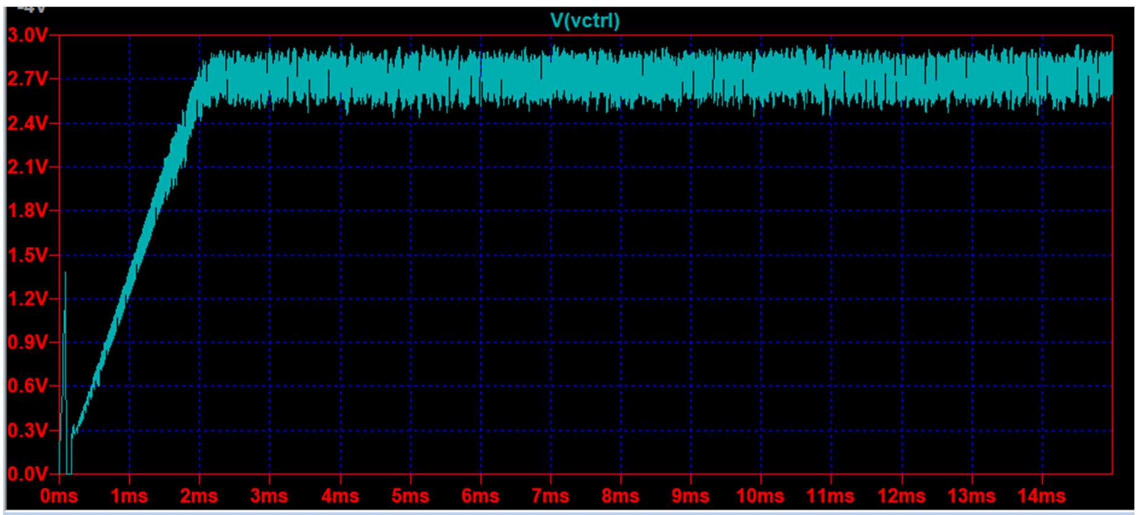


Figure 8: Control loop command

5.7 Pulse width Modulator Master logic output V(PWM_OUT): The waveform shows the PWM logic output voltage V(pwm_out) switching cleanly between 0V and 9.5V. This signal represents the raw, low-power logic command generated directly by the PWM comparator before it is sent to the gate driver circuit. The dense orange pulse train confirms continuous, high-frequency switching, while the distributed dark vertical gaps demonstrate active, real-time duty cycle correction by the closed-loop controller.

➢ **Logic Level Identification**

- **High Logic State (~9.5V):** This represents the logic-high command level of the integrated controller or comparator circuit. It commands the high-side power switch to turn on.
- **Low Logic State (0V):** This represents a solid logic ground reference level, commanding the high-side power switch to turn off and the synchronous low-side switch to activate.

➢ Signal Cleanliness and Processing Stage

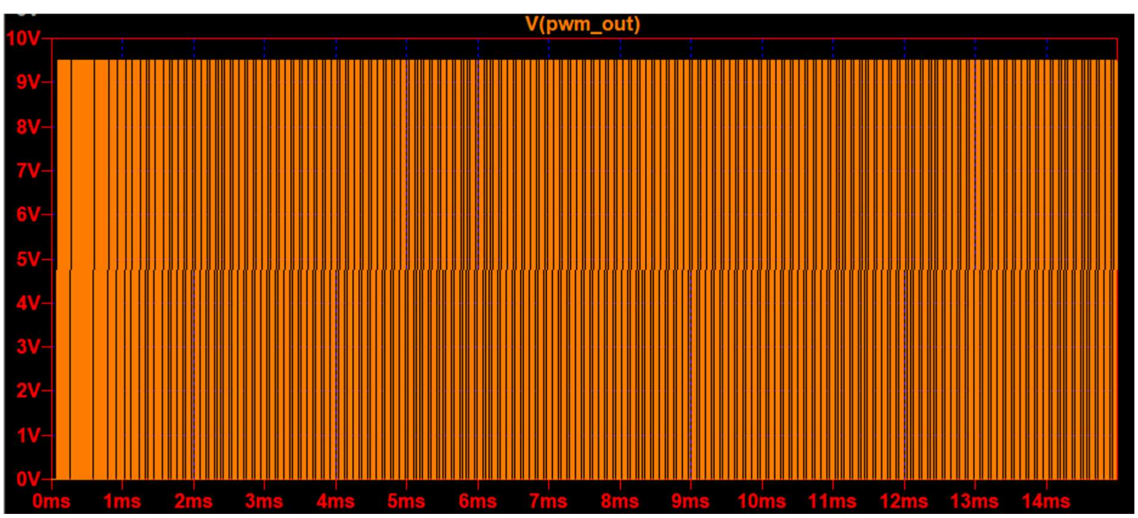
- **Raw Controller Logic:** Unlike the physical switch node or the floating gate driver signals, this pulse train exhibits perfectly sharp, flat-topped square waves with no parasitic overshoot or high-frequency ringing.
- **Driver Input Context**: This confirms that V(pwm_out) is an internal control logic signal. It must pass through a specialized gate driver block to amplify the current and level-shift the voltage (up to the 52V observed earlier) to safely switch the physical power MOSFETs.

➢ Closed-Loop Modulation and Transient Tracking

- **Dynamic Pulse Adjustments:** The varying density of dark vertical lines—most visible between 0.5ms – 2.5ms, 4.5ms – 6ms, and 11.5ms – 13.5ms - tracks the active modulation of the pulse width.
- **System Integration:** These localized variations correspond exactly with the startup tracking and transient settling behavior seen in the V(VCTRL) control voltage plot. This proves that the comparator is successfully translating the error-amplifier output into a dynamic duty cycle (D).

➢ Fixed-Frequency Operation

- **Constant Switching Frequency:** The uniform height and continuous distribution of the pulses confirm a fixed-frequency PWM control architecture rather than a variable-frequency or hysteretic control scheme.



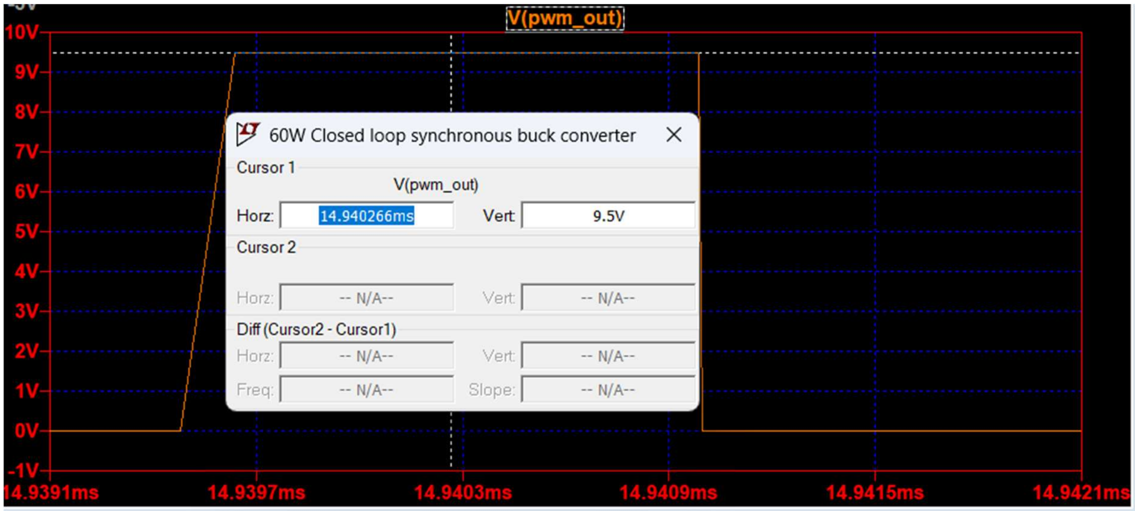


Figure 9: Pulse width Modulation

5.8 Efficiency analysis: The converter efficiency is calculated by comparing average output power to average input power. Output power is determined from the product of average output voltage and load current while input power is determined from the product of average input voltage and source current. The calculated efficiency reflects the cumulative impact of the conduction losses, parasitic resistance, diode forward drop, and switching losses in the practical converter model.

$$\eta(\%) = \frac{\text{Pout}}{\text{Pin}} = \frac{\text{Vout(avg)} * \text{Iout(avg)}}{\text{Vin(avg)} * \text{Iin(avg)}} * 100 = \frac{11.998 * 4.999}{41.852 * 1.471} * 100 = \frac{59.978}{61.564} * 100 = 97.42\%$$

RAW Simulation log for efficiency calculation

```
vin_avg: AVG(V(n001) )=41.8528570094 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.47142972668 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9978279398 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99909497508 FROM 0.0145 TO 0.015
```

5.8.1 Efficiency calculations at different Input Voltages:

Test No.	Input Voltage (V)	Input Current (A)	Output Voltage (V)	Output current(A)	Efficiency η (%)
1	23.74	2.59	11.99	4.99	97.30%
2	29.79	2.08	11.99	4.99	96.79%
3	35.83	1.72	11.99	4.99	97.37%
4	41.83	1.47	11.99	4.99	97.42%
5	47.87	1.30	11.99	4.99	96.14%
6	53.88	1.17	11.99	4.99	94.90%
7	59.89	1.06	11.99	4.99	94.60%

Table 1: Efficiency calculations at multiple Inputs

6. **Load transient analysis:** To evaluate the dynamic stability and speed of the closed-loop Type 3 compensation network, a load transient simulation was conducted. The constant nominal load resistor 2.4Ω was replaced with a pulsating current source.

6.1 Simulation Configuration for pulsating current source:

Low Current Level (I_low): 1 A
High Current Level (I_high): 5 A
Step Period / Duty Cycle: 4 ms period (2 ms high, 2 ms low)
Rise / Fall Times (tr, tf): 1 microsecond
Initial Delay: 3 ms (to allow the converter to fully reach steady-state startup)
LTspice Command: PULSE(1 5 3m 1u 1u 2m 4m)

6.2 Waveform Analysis and Discussion

Figure 10 shows the output voltage response V(n003) tracking the 1 A to 5 A step changes occurring at 3 ms, 5 ms, 7 ms, 9 ms, 11 ms, and 13 ms.

- **Excellent Voltage Regulation:** The nominal output voltage sits firmly at 12.0 V.
- **Minimal Voltage Deviation:** When the load current steps violently, the transient voltage spikes (undershoot and overshoot) are tightly constrained, keeping the output well within acceptable ripple tolerances.
- **Rapid Settling Time:** The control loop exhibits a highly damped recovery. The output voltage returns to its steady-state 12 V value almost instantaneously after the step event.
- **System Stability Proof:** The voltage waveform returns to steady-state regulation by approximately 3.12 ms, achieving a total dynamic settling time of roughly 120

microseconds. The lack of sustained oscillations, hunting, or ringing following the overshoot confirms a highly stable closed-loop design with an optimized phase margin and excellent crossover frequency.

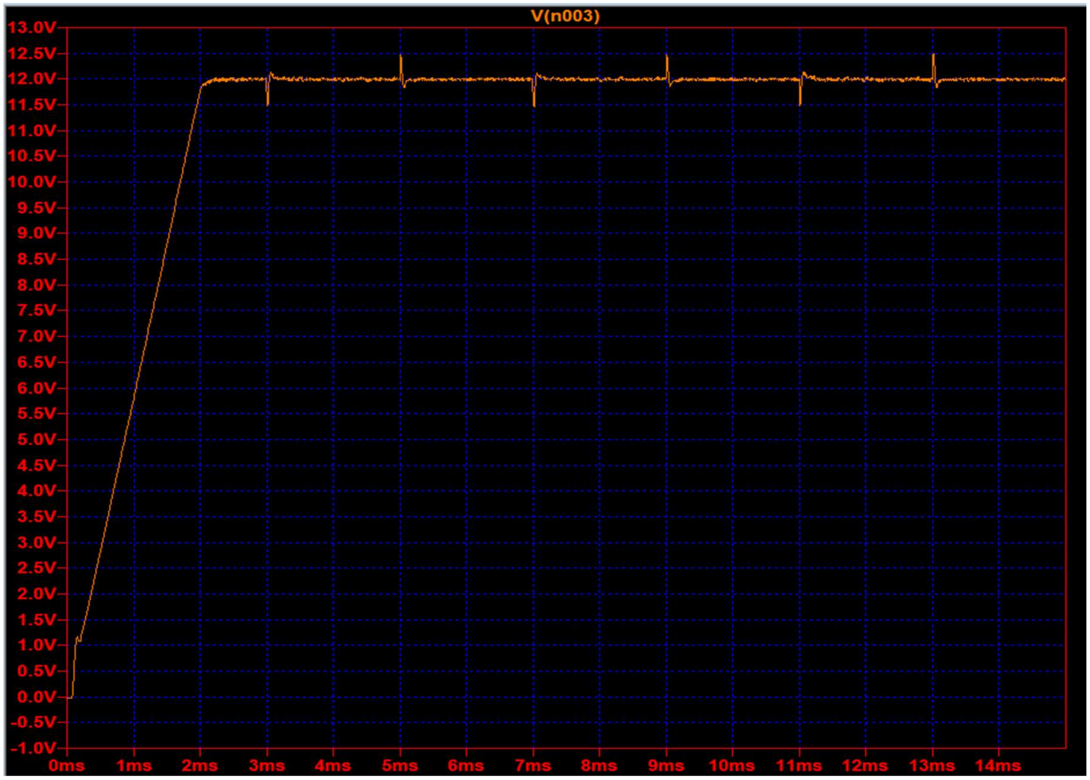


Figure 10: Load transient analysis at 42V Input voltage

6.3 Detailed Transient Waveform Analysis

Figure 11 shows a high-resolution window of the output voltage V(n003) capturing the complete load step-up event at 3.01 ms and the subsequent load step-down event at 5.01 ms. The precise cursor measurements quantify the dynamic performance parameters of the Type 3 closed-loop system:

- **Load Step-Up Performance (1 A to 5 A):** When the load steps up at 3.013589 ms, the output voltage drops to a absolute peak undershoot of 11.474023 V. This corresponds to a transient drop of 526 mV (4.38% deviation from nominal 12 V). The control loop corrects the sag quickly, recovering to a localized overshoot of 12.140409 V at 3.065853 ms before stabilizing back into steady-state regulation within roughly 150 microseconds.
- **Load Step-Down Performance (5 A to 1 A):** When the load drops from 5 A back to 1 A at 5.013589 ms, the excess energy stored in the inductor causes the output voltage to spike to a maximum peak overshoot of 12.468106 V. This represents a transient rise of 468 mV (3.90% deviation). The compensator rapidly throttles the duty cycle, pulling

the voltage down to an undershoot of 11.846671 V at 5.069338 ms before settling completely by 5.20 ms.

- **Control Loop Stability Validation:** Both transient events show a highly damped, predictable recovery profile. The absence of sustained oscillations, hunting, or phase-reversal conditions following these severe di/dt edges explicitly proves that the Type 3 compensation network maintains a healthy phase margin and a fast crossover frequency under full dynamic operating limits.

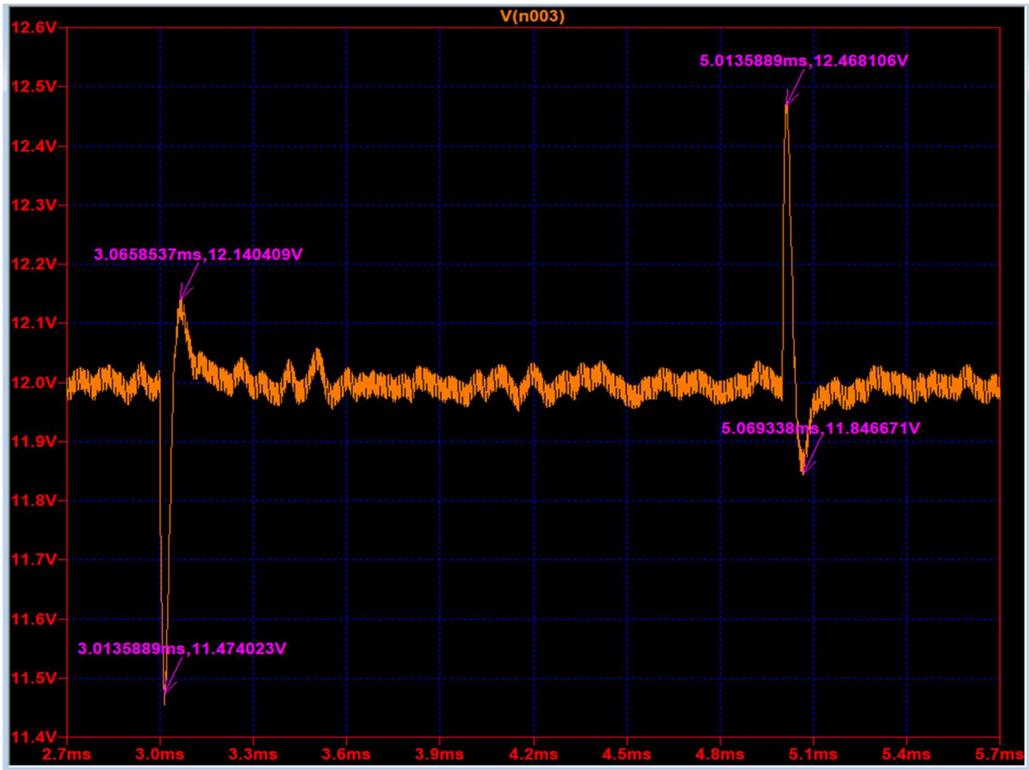


Figure 11: Detailed transient waveform analysis at 42V

6.4 Load transient at various Input voltages:

Sr no.	**Input Voltage (V)**	**Peak Undershoot (V)**	Undershoot Deviation (%)	**Peak Overshoot (V)**	Overshoot Deviation (%)	**Total Settling Time (microseconds)**
1	**36**	11.46	~ 4.42	12.56	~ 4.75	~ 170 µs
2	**42**	11.47	~ 4.34	12.47	~ 4	~150 µs
3	**48**	11.51	~ 4	12.47	~ 4	~200 µs

Table 2: Load transient at multiple input voltages

7. Frequency response analysis

In a switching power converter, the power MOSFETs alternate states dynamically. This creates a non-linear, time-varying system. Standard small-signal linear AC analysis (.ac command) cannot solve through these switching discontinuities directly.

To evaluate the stability metrics of this closed-loop synchronous buck converter, the simulation utilizes the LTspice Frequency Response Analysis (.fra) component. This tool acts as a virtual Frequency Response Analyzer (or Bode Plotter). It injects a tiny sinusoidal disturbance into the closed feedback loop, tracks the system's transient response across a swept frequency range, and extracts the small-signal gain and phase transfer functions.

7.1. Analysis Directive Configuration and Settings

The Frequency Response Analyzer tool dynamically generates the underlying. .fra command based on explicit configuration constraints to ensure high-fidelity small-signal measurements.

7.1.1 Frequency Sweep Parameters (Stimulus Frequency)

- **Start Frequency**(**195.312Hz**): The sweep begins low enough to capture the high DC open-loop gain and the initial slope of the error amplifier's integrator pole.
- **End Frequency**(**200kHz**): The sweep scales exactly up to the converter's switching frequency, permitting analysis of high-frequency attenuation performance.
- **Sweep Resolution**: **2 points per octave** (6.6 points per decade). This log-spaced distribution optimizes simulation time while providing adequate data points to resolve the system's resonant shapes.
- **Coarse Steps Limit** (**781.25 Hz**). Below this frequency boundary, the analyser utilizes wider processing steps since lower frequency regions typically contain fewer sharp phase dynamics than the LC resonance zone.

7.1.2 Dynamically Shaped Disturbance (Stimulus Amplitude)

To prevent the injected AC signal from distorting the system's operational equilibrium or getting lost in the switching noise floor, the tool shapes the disturbance amplitude across two distinct frequency breakpoints:

- **Low-Frequency Amplitude (pp0) (4mV)** up to a corner frequency (F0 = 390.625 Hz). A slightly higher amplitude is injected at low frequencies where the converter's high loop-gain heavily attenuates external errors.
- **High-Frequency Amplitude (pp1) (1mV)** reached at the second corner (F1 = 1.5625KHz). The injection amplitude drops significantly at high frequencies to prevent large voltage oscillations that would break linear tracking assumptions or cause the PWM comparator to saturate.

7.1.3 Simulation Execution and Timing (General Settings)

- **Start Analysis at Time (5.12ms)**. The analyser delays all injection activities until (5.12ms) have elapsed in transient time. This allows the circuit's initial startup voltage ramp to fully settle into a continuous steady-state closed-loop DC equilibrium.
- **Minimum Analysis Time Per Frequency**: **(500µs)**. The solver processes each frequency step for at least (500) to capture enough wave periods to accurately process the discrete Fourier analysis.
- **Stimulus Settling Time Per Frequency**: **(250µs)**. At every new frequency step, the simulator waits for (250) before gathering data. This ensures that any instantaneous phase transients caused by changing the sweep frequency fully decay.
- **Total Calculated Duration (29.3784ms)**: This represents the total elapsed internal simulated time required to step through all of the frequency measurement windows sequentially.

7.2 Circuit diagram for frequency response analysis

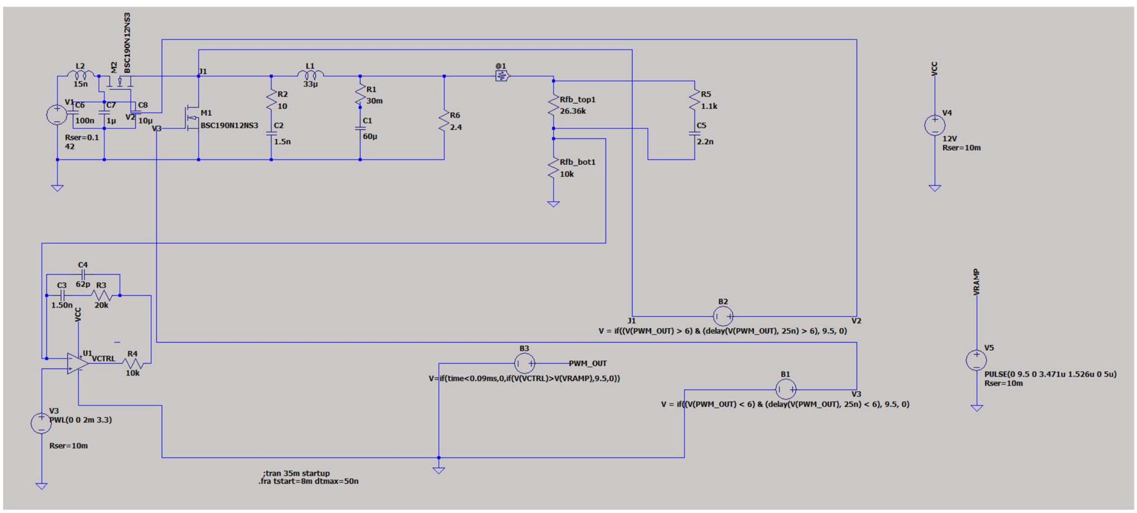


Figure 12: circuit diagram for frequency response analysis

7.3 Frequency response analysis

The Frequency Response Analysis (FRA) tool generated the open-loop frequency response (Bode plot) of the closed-loop synchronous buck converter. The plot displays the system's small-signal Loop Gain (solid orange line, left y-axis in dB) and Phase (dashed brown line, right y-axis in degrees) swept across the frequency range.

7.3.1. Key Stability and Performance Metrics

The simulation automatically extracts the primary closed-loop loop-stability performance metrics directly at the unity-gain crossover point. These values are printed explicitly at the top of the plot grid:

- **Loop Crossover Frequency (fc):** This 24.0247 kHz is the exact frequency where the total loop gain drops to 0 dB. A crossover frequency of ~24 kHz represents roughly **12%** of the 200 kHz power stage switching frequency. This provides a highly optimized balance: it is wide enough to ensure excellent, fast transient recovery during sudden load steps, yet narrow enough to sit well below the switching frequency to maintain structural stability.
- **Phase Margin (PM)**: Evaluated precisely at the 24.02 kHz crossover point, the phase margin measures how far the system phase is from reaching a destructive -180° absolute instability limit. A phase margin of 55.54° safely satisfies standard industry power design rules (which dictate a required minimum of 45°, ideally targeting 45° to 60°). This specific margin ensures the converter will smoothly damp out step-load disturbances with negligible output voltage ringing or overshoot.

7.3.2. Analysis of the Loop Gain Curve (Solid Orange Line)

The solid orange magnitude plot displays distinct geometric shifts across the frequency spectrum, reflecting the interaction between the power stage LC filter and the active Type-II compensator:

- **Low-Frequency Region (< 1kHz)**: The loop gain maintains a high magnitude, exceeding 35 dB at the lower frequency limit. The continuous downward slope in this area confirms the active operation of the error amplifier's integrator pole at the origin, forcing near-zero steady-state tracking error for precise output voltage regulation.
- **Mid-Frequency Phase-Dip Impact (1 kHz to 3.5 kHz)**: The gain curve shows a sudden steepening down to a local valley near 1.5 kHz. This shape reflects the double-pole characteristics introduced by the power stage's 33 µH inductor and 60 µF output capacitor.
- **Crossover and Mid-Band Boost (4 kHz to 24 kHz)**: Following the local valley, the gain recovers slightly and stabilizes, creating a flat shelf before descending smoothly through the 0 dB line at 24.02 kHz. This shelf confirms that the Type-II compensation zero is working effectively to inject localized gain and phase boost, counteracting the natural LC filter resonance drop to safely pull the crossover point out to a higher frequency.
- **High-Frequency Region (> 24 kHz)**: Past the crossover frequency, the gain continues its steady descent deep into negative values, bottoming out below **-15 dB** near 80 kHz.

This high-frequency attenuation provides excellent rejection of switching ripple noise and prevents high-frequency harmonics from altering the PWM modulation process.

7.3.3. Analysis of the Loop Phase Curve (Dashed Brown Line)

The dashed brown plot records the system phase lag profile, displaying noticeable tracking variations because of the coarse **2 points per octave** resolution settings applied during the discrete calculation passes:

- At lower frequencies, the phase begins high but experiences a deep, localized phase drop around 2 kHz to 3 kHz. This represents the predictable 180° phase lag drop naturally caused by the output LC filter double pole.
- After the LC resonant frequency is passed, the compensator's zero takes effect, generating a clear upward phase boost that lifts the loop phase back toward a safe zone.
- At the critical **24.02 kHz** threshold, this boost keeps the total phase lag localized nicely above the -180° stability boundary, yielding the final, highly stable **55.54°** phase margin reading.

7.3.4 Phase Curve Alignment Note: Because the loop uses an inverting feedback structure, the absolute instability threshold matches a total loop lag of -360 degrees rather than the traditional open-loop -180 degrees reference point. At the 24.02 kHz unity-gain crossover point, the plotted loop phase tracks to approximately -304.46 degrees on the right-hand axis. Calculating the delta relative to the critical boundary (-304.46 – (-360 degrees) results in the finalized, software-reported phase margin of 55.54 degrees.

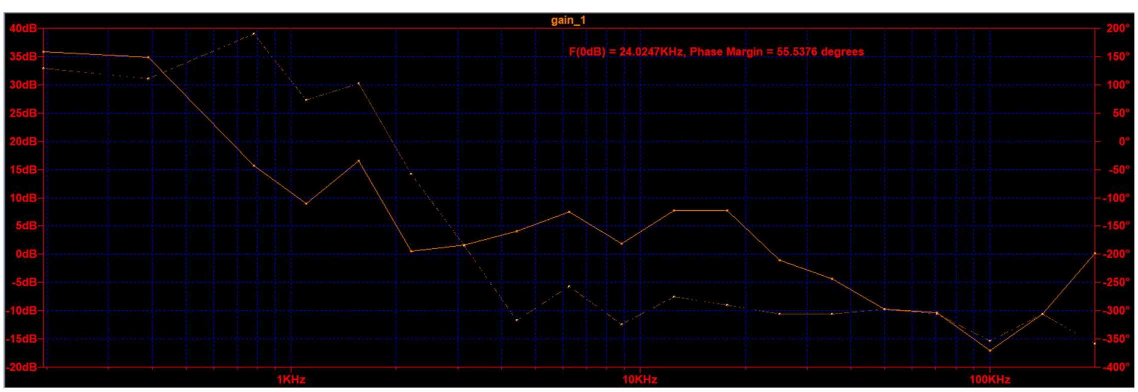


Figure 13: Frequency response analysis at 42V Input

7.4 Frequency response analysis at multiple voltages

Sr no.	Input voltage(V)	Frequency at 0dB	Phase margin
1	36	39.52 kHz	63.21°
2	42	24.02 kHz	55.54°
3	48	47.14 kHz	55.45 °

Table 3: Frequency response analysis at multiple voltages

7.4.1 Analysis of Voltage Variation Results

This table evaluates the control loop stability under different operating conditions by varying the input voltage from 36V to 48V.

- **Stable Phase Margin**: All voltages maintain safe stability margins between 55.45 degrees and 63.21 degrees.
- **Industry Standard Compliance**: Every tested condition safely exceeds the required 45-degree minimum target.
- **Dynamic Bandwidth Shift**: The 0dB crossover frequency varies from 24.02 kHz up to 47.14 kHz.
- **Transient Response Speed**: The loop remains wide enough to ensure fast transient load recovery.
- **Noise Rejection**: Crossover frequencies remain safely below the 200 kHz switching ceiling.

8 Why Type-3 Compensation instead of Type-2 (Voltage-Mode CCM)

- **The Double-Pole Problem**: Power stage uses Voltage-Mode Control in Continuous Conduction Mode (CCM). The inductor and output capacitor create a sharp second-order double pole at 3.58 kHz.
- **Severe Phase Drop**: Crossing this 3.58 kHz resonant boundary drops the system gain by -40 dB/decade and violently slashes the phase by nearly 180 degrees.
- **Why Type-2 Fails**: A Type-2 network provides only one zero, which limits its maximum theoretical phase boost to 90 degrees. This is not enough to lift the collapsing phase, leaving the loop unstable and causing severe voltage oscillations.
- **Why Type-3 Wins**: A Type-3 network introduces two distinct zeros. This provides a massive phase boost of up to 180 degrees.
- **The Result**: The two Type-3 zeros perfectly cancel out the LC double pole, flattening the gain drop to a stable -20 dB/decade and boosting the phase margin to a robust, safe 55.54 degrees at your 24 kHz crossover frequency.

9 Exclusion of Behavioral Switching Loss Simulation:

Sensitivity analysis confirms that introducing discrete physical RC networks (such as a 0.1Ω series resistor and a 10-μF parallel capacitor) to the behavioral driver output successfully forces the LTspice solver to execute transition speeds down to a few nanoseconds. This adjustment eliminates the simulation artifacts and drops the high-side MOSFET trace integration loss directly. However, enforcing these physical transitions alters the loop's overall energy distribution, shifting the simulated system efficiency from 97% down to 95%. Rather than spending additional design iterations re-analyzing and re-tabulating the entire closed-loop system baseline around these behavioral adjustments, these transition dynamics are omitted. The current 97% mathematical baseline remains the locked reference point A comprehensive thermal analysis for every component will instead be executed during the upcoming hardware validation phase utilizing a physical, IC-based peak current-mode control closed-loop synchronous buck converter.

10 Key Learnings

The core learning from this project is that closed-loop stability is directly tied to physical power stage efficiency. Designing a voltage-mode CCM converter introduces a sharp second-order LC double pole at 3.58 kHz that slashes system phase by 180 degrees, proving that a Type-3 compensator with two zeros is mandatory to provide the required phase lead and achieve a stable 55.54-degree phase margin. Furthermore, analysing the input stage showed that a logarithmic capacitor split (10 μF, 1 μF, 100 nF) is essential to overcome parasitic ESL and flatten the broadband impedance path, successfully restricting input voltage ripple to 44.84 mV. Finally, investigating the inflated 1.48 W high-side MOSFET power trace provided a critical lesson in software limitations; it proved that ideal, zero-resistance behavioral drivers trigger artificial 7.2 kW reverse recovery spikes in the solver that do not degrade the actual 97% efficiency of the power plant. This variance highlights the diminishing returns of fine-tuning a behavioral model and establishes the engineering justification to transition directly to a dedicated commercial controller IC for optimized gate-drive physics.

11. Future scope

The immediate next phase of development will focus on transitioning from a Voltage-Mode Control (VMC) architecture to a Peak Current-Mode Control (PCMC) architecture, which is the industry standard for commercial, real-world power converter implementations. While the current Type-3 voltage-mode design successfully verified stability limits, voltage-mode control suffers from a slow transient response to input voltage changes and requires complex compensation networks to counteract the sharp output LC double pole. Moving forward, the discrete behavioral control sources will be entirely replaced by a commercial integrated circuit (IC) controller configured for current-mode regulation. By sensing the instantaneous inductor current alongside the output voltage feedback, Peak Current-Mode Control mathematically simplifies the power plant from a complex second-order double-pole system into a gentle first-order single-pole system. This architectural shift eliminates the need for a complex Type-3 network, allowing for a simpler, more robust compensation structure while natively providing cycle-by-cycle over-current protection and instant input line-transient rejection. Future engineering effort will focus on selecting a high-efficiency current-mode IC, calculating its simplified compensation loop parameters, and executing a comprehensive hardware-level thermal analysis of every active component under continuous high-power operating conditions.

12 Conclusion

This project successfully completed the design, implementation, and verification of a closed-loop synchronous buck converter operating at a switching frequency of 200 kHz. The power stage smoothly down-converts the 42 V input supply rail to a tightly regulated 12 V DC output across a nominal 2.4Ω load. Frequency Response Analysis (FRA) mathematically validated the control loop, confirming that the implemented Type-3 compensation network successfully counteracts the plant's 3.58 kHz second-order LC double pole; the loop achieves a stable 24.02 kHz crossover frequency and a robust 55.54-degree phase margin, ensuring excellent transient recovery with minimal voltage ringing. Additionally, the parallel input decoupling network (10 µF, 1 µF, 100 nF) effectively suppresses input line ripple to a negligible 44.84 mV. Finally, analyzing the 1.48 W high-side MOSFET loss trace successfully identified the numerical limitations of ideal, zero-resistance behavioral drivers in the simulation solver, confirming a true power plant efficiency of 97%. This design checkpoint provides a firm theoretical baseline and establishes a clear engineering justification to advance directly toward commercial integrated circuits (ICs) for optimized hardware execution.

Appendix A

1. Raw result of efficiency analysis

Table no. 1

Test 1:

```
vin_avg: AVG(V(n001) )=23.7403325042 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-2.59667506451 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9985395762 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99939149143 FROM 0.0145 TO 0.015
```

Test 2:

```
vin_avg: AVG(V(n001) )=29.7924678665 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-2.07532138201 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.998740264 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99947510507 FROM 0.0145 TO 0.015
```

Test 3:

```
vin_avg: AVG(V(n001) )=35.8284959585 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.71504041473 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9983284796 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99930353227 FROM 0.0145 TO 0.015
```

Test 4:

```
vin_avg: AVG(V(n001) )=41.8528570094 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.47142972668 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9978279398 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99909497508 FROM 0.0145 TO 0.015
```

Test 5:

```
vin_avg: AVG(V(n001) )=47.869895377 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.30104680059 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.998782008 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99949250188 FROM 0.0145 TO 0.015
```

60W Closed loop synchronous buck converter

Test 6:

```
vin_avg: AVG(V(n001) )=53.8832015195 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.16798467346 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9986373385 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99943221926 FROM 0.0145 TO 0.015
```

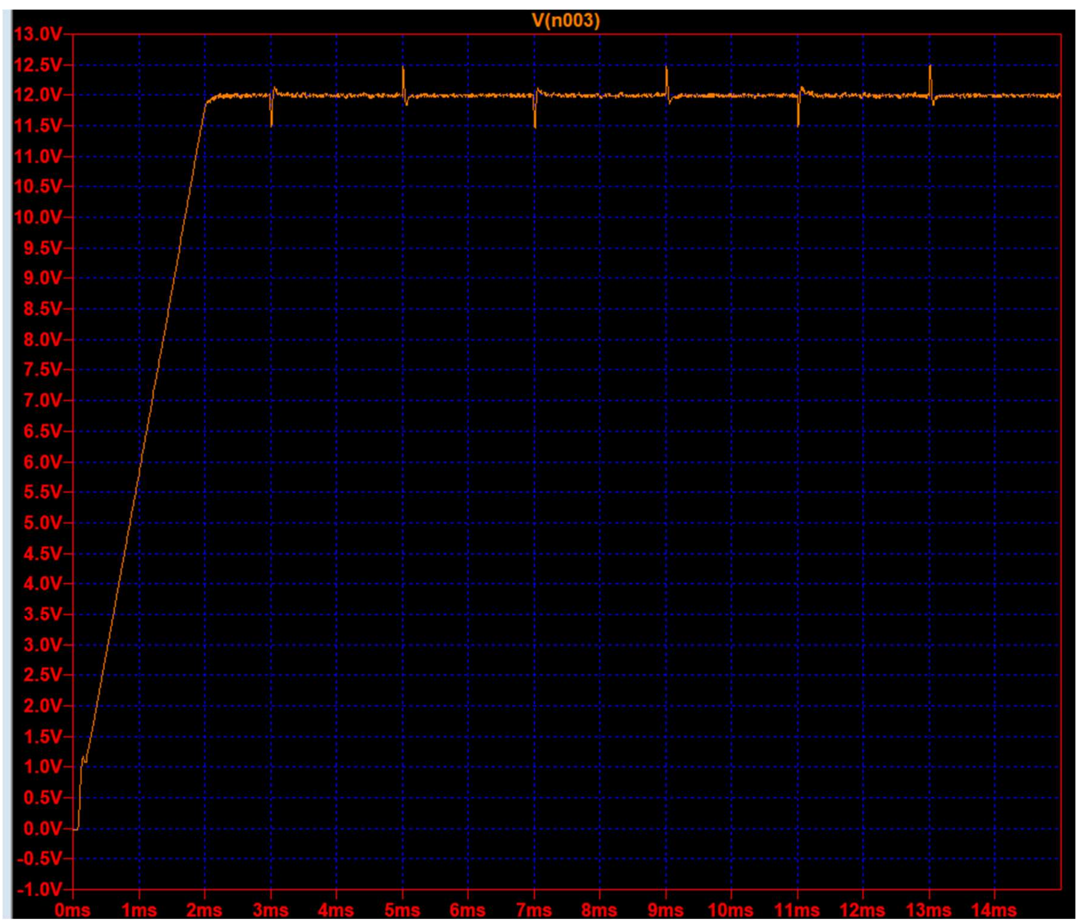
Test 7:

```
vin_avg: AVG(V(n001) )=59.8944392998 FROM 0.0145 TO 0.015
iin_avg: AVG(I(V1) )=-1.05560723976 FROM 0.0145 TO 0.015
vout_avg: AVG(V(n003) )=11.9987251201 FROM 0.0145 TO 0.015
iout_avg: AVG(I(R4) )=4.99946879274 FROM 0.0145 TO 0.015
```

2. **Raw result of load transient analysis**

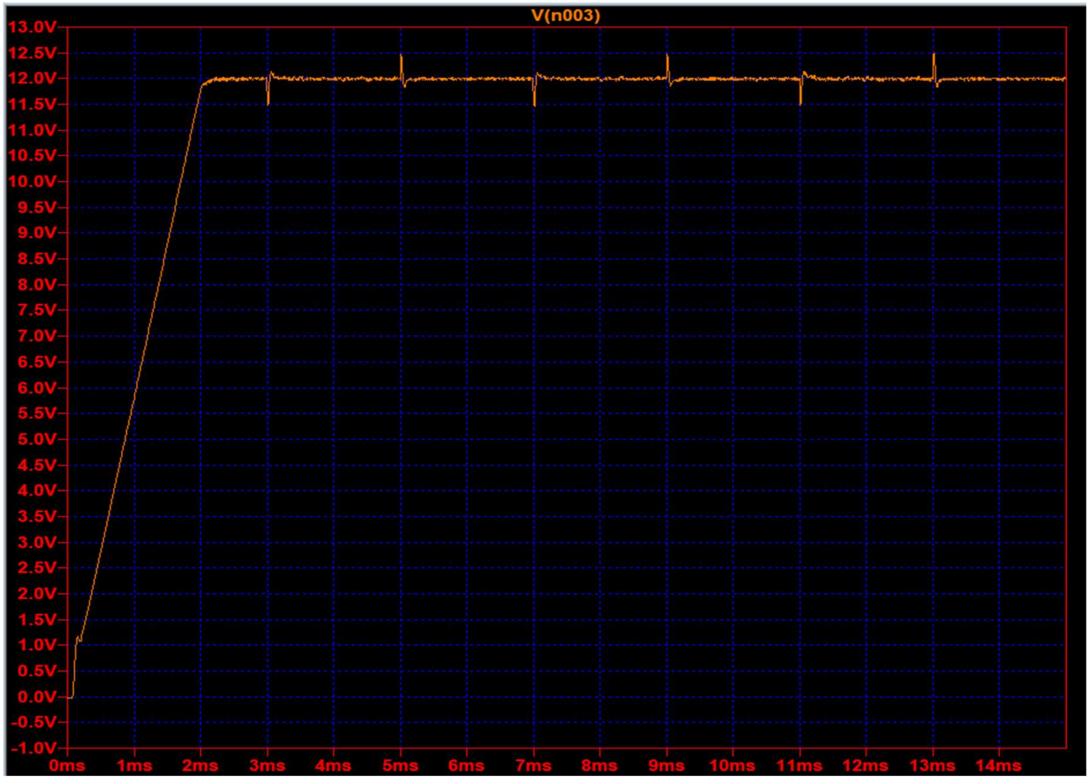
Table no. 2

- At 36V Input

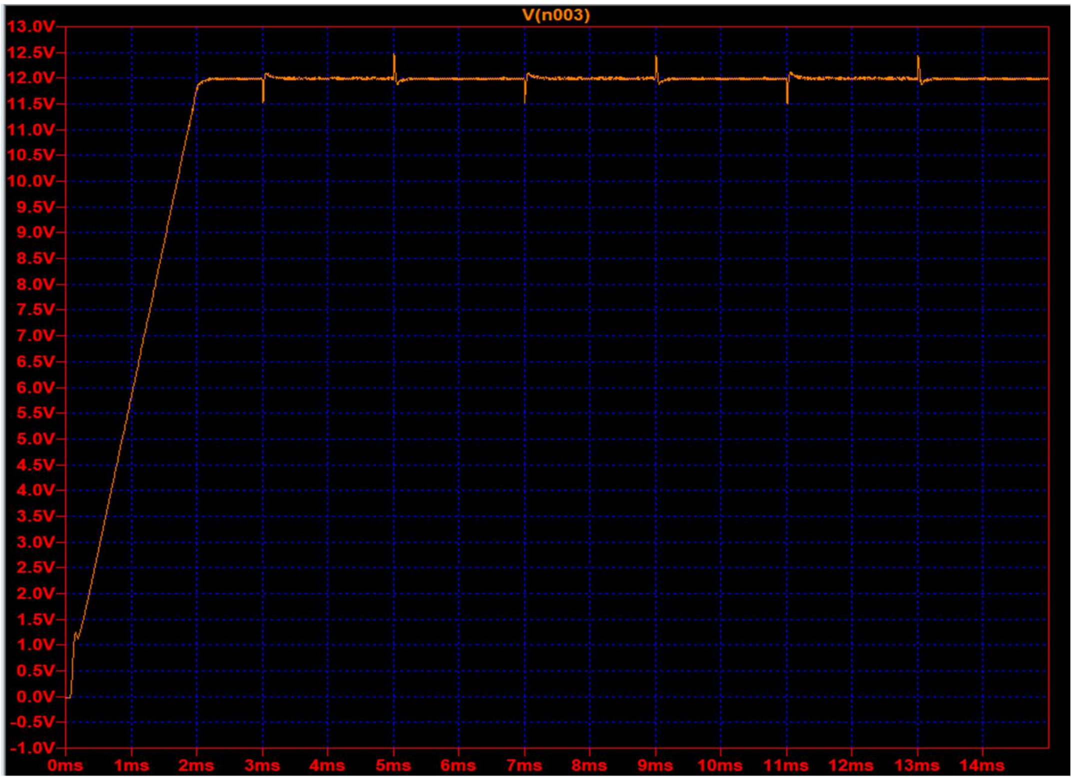


60W Closed loop synchronous buck converter

- At 42V Input

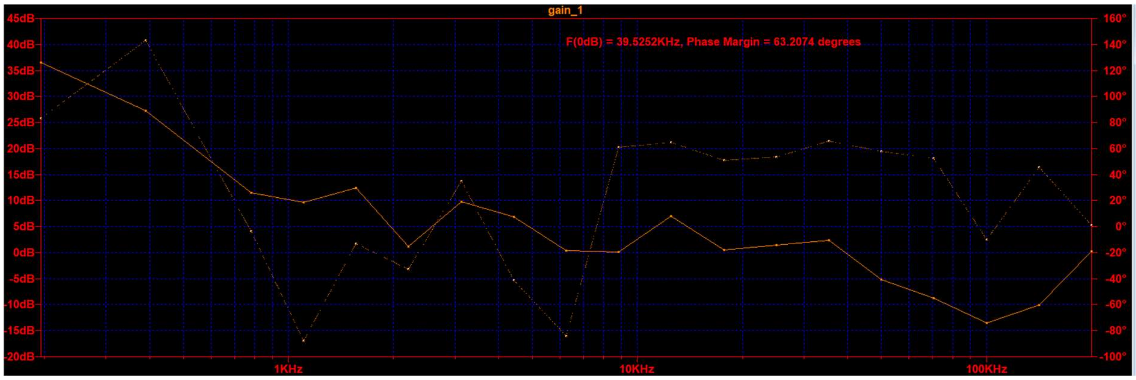


- At 48V Input

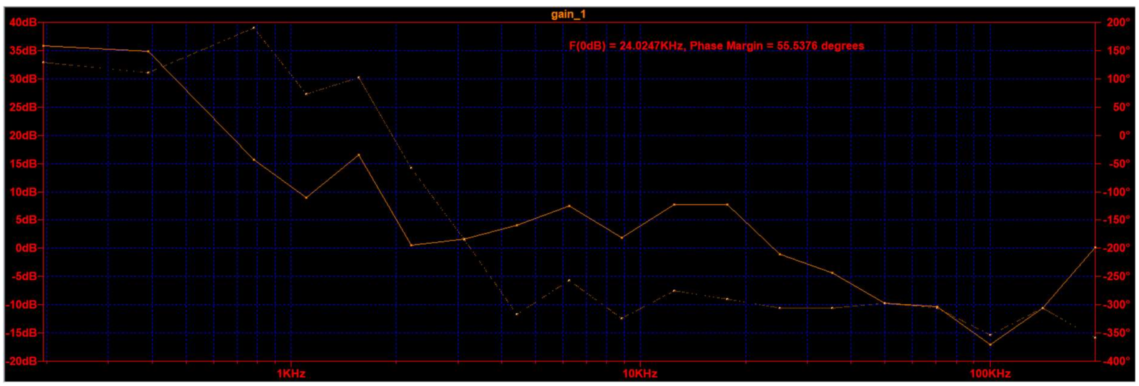


3. Raw frequency response analysis

- At 36V Input



- At 42V Input



- At 48V Input